

# NYCU-EE IC LAB – 2024 Fall

## Lab12 Check List

### Self-Verify APR Result

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.f02\_check under 09\_SUMBIT to download your Lab12\_iclabXXX.tar file back  
Create a new directory, enter the directory and decompress the tar file.  
Enter the decompressed directory.

1. Make sure your CHIP\_iclabXXX.sdc is written correctly: period, waveform parameter, input delay and output delay. Waveform parameter, input delay and output delay should be half of the period.
2. Invoke innonus and restore CHIP\_iclabXXX.inn  
( Remember to create a new folder in case you overwrite previous design)
3. Explore the core size and die size, also verify if the core to IO boundary should be larger than 100.
4. Verifying if the IO Filler and the corner pad is added.
5. Verify the floorplan and powerplan constraints:
  - a. Power ring: wire group, interleaving, and at least 4 pairs, width 9.
6. Post-Route Timing analysis with non-negative slacks, 0 DRV's, core filler added.
7. Verifying Geometry and Connectivity after adding core filler cells.
8. Latency cycles in post simulation should be the same as gate level simulation.